

重回帰分析のモデル比較 2

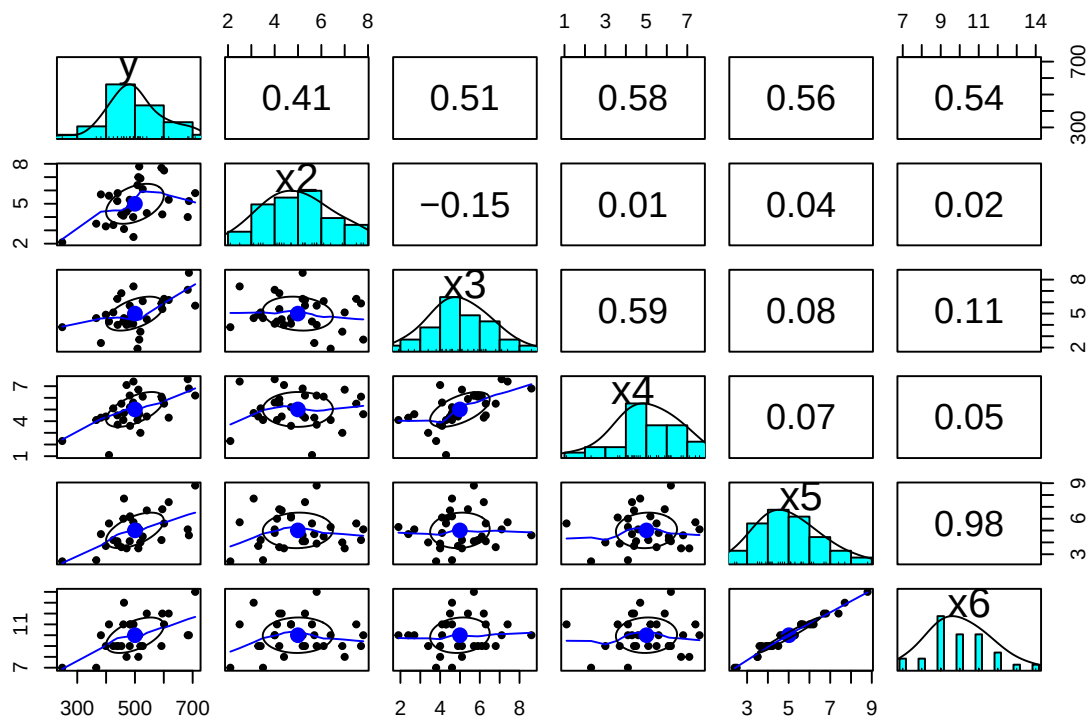
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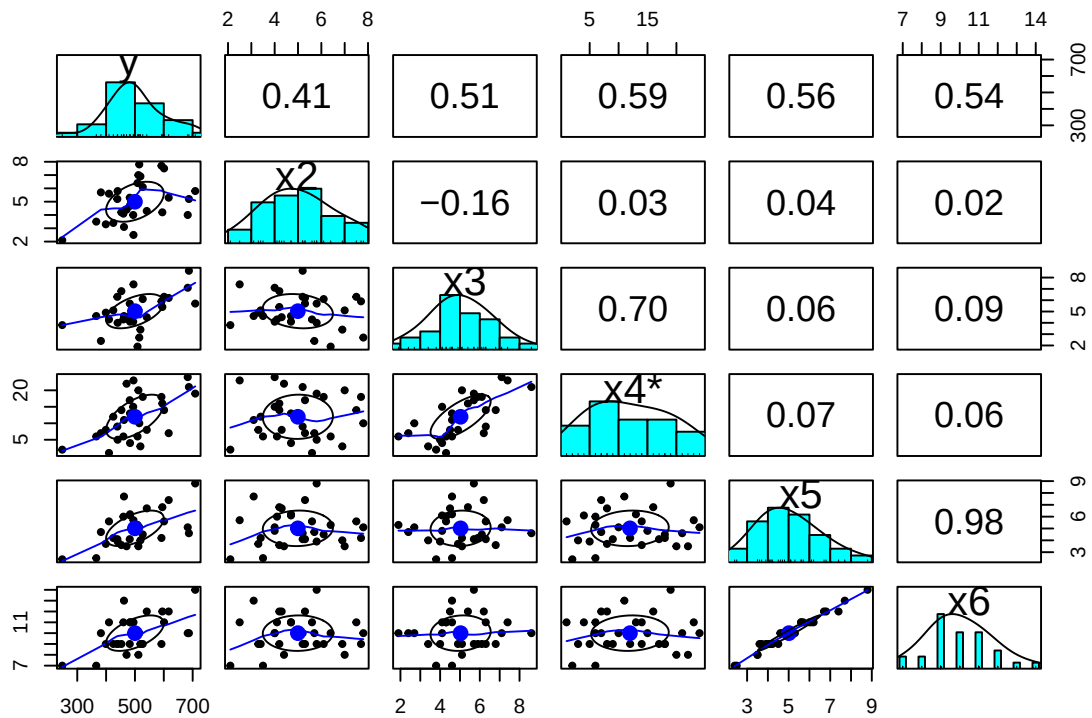
```
# ライブラリの読み込み
# install.packages("psych") # 初回のみ実行
library(psych)

# データの読み込み
cs <- read.csv("CS.csv")
cs2 <- read.csv("CS2.csv", header=TRUE, fileEncoding="cp932")

csdata <- cs[2:7]
psych::pairs.panels(csdata)
```



```
cs2data <- cs2[2:7]
psych::pairs.panels(cs2data)
```



```
# 平均値の確認
```

```
mean(cs2$x2)
```

```
## [1] 4.993333
```

```
mean(cs2$x3)
```

```
## [1] NA
```

```
mean(cs2$x3, na.rm=TRUE)
```

```
## [1] 5.031034
```

```
mean(cs2$x4)
```

```
## [1] NA
```

```
# クロス集計表
```

```
print("Table: CS x3")
```

```
## [1] "Table: CS x3"
```

```
table(cs$x3)
```

```
##
```

```
## 1.9 2.4 2.7 3.4 3.8 4 4.1 4.3 4.5 4.6 4.9 5.1 5.4 5.7 5.9 6.1 6.2 6.3 6.8 7.1
## 1 1 1 1 1 2 2 2 1 3 1 2 1 2 1 1 1 2 1 1
## 7.4 8.6
## 1 1
```

```
print("Table: CS x4")
```

```
## [1] "Table: CS x4"
```

```
table(cs$x4)
```

```
##
```

```
## 1.1 2.3 3 3.6 3.7 4.1 4.3 4.4 4.5 4.6 4.7 5.1 5.2 5.5 5.6 5.9 6.1 6.2 6.7 6.8
## 1 1 1 1 1 3 2 2 1 1 1 1 1 2 1 1 2 2 1 1
## 7.1 7.4 7.6
## 1 1 1
```

```
print("Table: CS2 x4")
```

```
## [1] "Table: CS2 x4"
```

```
table(cs2$x4)
```

```
##
```

```
## 1.1 2.3 3 3.6 3.7 4.1 4.3 4.4 4.5 4.6 4.7 5.1 5.2 5.5 5.6 5.9 6,2 6.1 6.2 6.7
## 1 1 1 1 1 3 2 2 1 1 1 1 1 2 1 1 1 2 1 1
## 6.8 7.1 7.4 7.6
## 1 1 1 1
```

```
print("Table: CS2 x7")
```

```
## [1] "Table: CS2 x7"
```

```
table(cs2$x7)
```

```
##
```

```
## 0 1 Y
## 15 13 2
```

```
print("Table: CS2 x8")
```

```
## [1] "Table: CS2 x8"
```

```
table(cs2$x8)
```

```
##  
##  0 0  1 1  
## 17  1 11  1
```

```
#model1
```

```
# CS データでの分析
```

```
fit1_cs <- lm(y ~ x2 + x3 + x4 + x5, data=cs)  
summary(fit1_cs)
```

```
##  
## Call:  
## lm(formula = y ~ x2 + x3 + x4 + x5, data = cs)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -63.636 -36.548  -0.314   21.419  101.890   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -35.204     50.659  -0.695  0.49351      
## x2             29.105      5.684   5.120 2.73e-05 ***   
## x3             21.640      7.036   3.076 0.00503 **    
## x4             23.803      6.971   3.414 0.00219 **    
## x5             32.421      5.608   5.781 5.01e-06 ***   
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 45.11 on 25 degrees of freedom  
## Multiple R-squared:  0.8248, Adjusted R-squared:  0.7968   
## F-statistic: 29.43 on 4 and 25 DF,  p-value: 3.957e-09
```

```
# CS2 データでの分析
```

```
fit1_cs2 <- lm(y ~ x2 + x3 + x4 + x5, data=cs2)  
summary(fit1_cs2)
```

```
##  
## Call:  
## lm(formula = y ~ x2 + x3 + x4 + x5, data = cs2)  
##
```

```
## Residuals:
##          1          2          3          4          5          6          7
## -2.423e-15 -3.686e-15 -2.180e+01  7.859e-17  1.520e+01  4.891e+00 -4.475e-15
##          8          9         11         12         13         14         15
##  6.440e-16 -5.079e-16 -7.516e-16  3.453e-15 -1.520e+01 -1.100e-16  2.180e+01
##         16         17         18         19         20         21         22
##  3.908e-16  3.260e-15  1.678e-15 -4.891e+00  9.527e+00  4.910e-15  1.322e-15
##         23         24         25         26         27         28         29
##  1.647e+00  2.819e-15 -2.846e+01 -3.111e-15  2.681e+01  8.067e-18 -9.527e+00
##         30
##  3.103e-15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -171.82     106.03  -1.621   0.2036
## x2              44.88      14.84   3.024   0.0566 .
## x3              36.61      13.20   2.774   0.0693 .
## x42.3           112.28      62.70   1.791   0.1713
## x43             133.05      53.06   2.507   0.0871 .
## x43.6           105.26      49.85   2.112   0.1252
## x43.7            84.54      49.41   1.711   0.1856
## x44.1           133.42      41.01   3.253   0.0474 *
## x44.3            80.08      40.26   1.989   0.1408
## x44.4           138.50      45.01   3.077   0.0543 .
## x44.5            20.08      56.51   0.355   0.7458
## x44.6            80.29      53.03   1.514   0.2272
## x44.7            88.32      69.17   1.277   0.2915
## x45.1           127.71      50.63   2.523   0.0860 .
## x45.2            83.87      48.60   1.726   0.1828
## x45.5            54.31      55.89   0.972   0.4029
## x45.6           187.86      50.57   3.714   0.0339 *
## x45.9           170.48      51.53   3.309   0.0454 *
## x46,2            80.56      53.16   1.516   0.2269
## x46.1            67.54      62.27   1.085   0.3575
## x46.2           139.88      53.82   2.599   0.0805 .
## x46.7            74.77      65.59   1.140   0.3370
## x46.8           168.43      74.56   2.259   0.1090
## x47.4           107.55      57.64   1.866   0.1589
## x47.6           257.75      53.83   4.788   0.0173 *
## x5              30.90      10.54   2.932   0.0609 .
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.53 on 3 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.989, Adjusted R-squared:  0.8977
## F-statistic: 10.83 on 25 and 3 DF, p-value: 0.03643
```

```
extractAIC(fit1_cs2)
```

```
## [1] 26.0000 188.1648
```

```
#model2
```

```
# CS データでの分析
```

```
fit2_cs <- lm(y ~ x2 + x3 + x4 + x5 + x7 + x8, data=cs)
summary(fit2_cs)
```

```
##
## Call:
## lm(formula = y ~ x2 + x3 + x4 + x5 + x7 + x8, data = cs)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
##	-78.543	-25.702	-2.392	17.573	91.820

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-13.878	48.906	-0.284	0.77913
## x2	27.488	5.544	4.958	5.18e-05 ***
## x3	24.745	6.779	3.650	0.00134 **
## x4	17.963	7.067	2.542	0.01822 *
## x5	28.515	5.592	5.099	3.65e-05 ***
## x7	8.014	16.121	0.497	0.62383
## x8	40.128	18.065	2.221	0.03646 *

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 42.55 on 23 degrees of freedom
## Multiple R-squared:  0.8566, Adjusted R-squared:  0.8192
## F-statistic: 22.9 on 6 and 23 DF, p-value: 1.27e-08
```

```
# CS2データでの分析
```

```
fit2_cs2 <- lm(y ~ x2 + x3 + x4 + x5 + x7 + x8, data=cs2)
```

```
summary(fit2_cs2)
```

```
##
```

```
## Call:
```

```
## lm(formula = y ~ x2 + x3 + x4 + x5 + x7 + x8, data = cs2)
```

```
##
```

```
## Residuals:
```

```
## ALL 29 residuals are 0: no residual degrees of freedom!
```

```
##
```

```
## Coefficients: (2 not defined because of singularities)
```

```
##           Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -117.74         NaN      NaN      NaN
```

```
## x2           -27.91         NaN      NaN      NaN
```

```
## x3           -22.42         NaN      NaN      NaN
```

```
## x42.3        234.86         NaN      NaN      NaN
```

```
## x43          447.75         NaN      NaN      NaN
```

```
## x43.6        125.68         NaN      NaN      NaN
```

```
## x43.7        362.95         NaN      NaN      NaN
```

```
## x44.1        101.86         NaN      NaN      NaN
```

```
## x44.3         35.20         NaN      NaN      NaN
```

```
## x44.4        112.81         NaN      NaN      NaN
```

```
## x44.5        234.35         NaN      NaN      NaN
```

```
## x44.6        187.77         NaN      NaN      NaN
```

```
## x44.7       -111.90         NaN      NaN      NaN
```

```
## x45.1        131.76         NaN      NaN      NaN
```

```
## x45.2        125.45         NaN      NaN      NaN
```

```
## x45.5        288.88         NaN      NaN      NaN
```

```
## x45.6        126.42         NaN      NaN      NaN
```

```
## x45.9        172.75         NaN      NaN      NaN
```

```
## x46,2        406.21         NaN      NaN      NaN
```

```
## x46.1        436.73         NaN      NaN      NaN
```

```
## x46.2       -30.28         NaN      NaN      NaN
```

```
## x46.7        538.89         NaN      NaN      NaN
```

```
## x46.8        616.21         NaN      NaN      NaN
```

```
## x47.4        196.03         NaN      NaN      NaN
```

```
## x47.6        191.29         NaN      NaN      NaN
```

```
## x5           114.45         NaN      NaN      NaN
```

```
## x71          139.52          NaN          NaN          NaN
## x7Y          296.57          NaN          NaN          NaN
## x8 0          182.42          NaN          NaN          NaN
## x81           NA           NA           NA           NA
## x8 1           NA           NA           NA           NA
##
## Residual standard error: NaN on 0 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:      1, Adjusted R-squared:      NaN
## F-statistic:      NaN on 28 and 0 DF, p-value: NA
```

```
#model3
```

```
# CS データでの分析
```

```
fit3_cs <- lm(y ~ x2 + x3 + x4 + x5 + x8, data=cs)
summary(fit3_cs)
```

```
##
## Call:
## lm(formula = y ~ x2 + x3 + x4 + x5 + x8, data = cs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -80.658 -23.207  -0.821  16.363  87.851
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -12.787     48.084  -0.266  0.79256
## x2             28.154       5.294   5.318 1.86e-05 ***
## x3             24.526       6.658   3.684 0.00117 **
## x4             18.203       6.939   2.623 0.01490 *
## x5             28.441       5.501   5.170 2.71e-05 ***
## x8             39.752      17.764   2.238 0.03478 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.88 on 24 degrees of freedom
## Multiple R-squared:  0.8551, Adjusted R-squared:  0.8249
## F-statistic: 28.32 on 5 and 24 DF, p-value: 2.515e-09
```

CS2 データでの分析

```
fit3_cs2 <- lm(y ~ x2 + x3 + x4 + x5 + x8, data=cs2)
summary(fit3_cs2)
```

```
##
## Call:
## lm(formula = y ~ x2 + x3 + x4 + x5 + x8, data = cs2)
##
## Residuals:
```

##	1	2	3	4	5	6	7
##	2.829e-16	-2.543e-15	-9.208e+00	-1.969e-15	1.341e+01	-1.215e+01	4.535e-15
##	8	9	11	12	13	14	15
##	7.007e-16	-2.999e-15	3.267e-15	-2.854e-15	-1.341e+01	-1.609e-15	9.208e+00
##	16	17	18	19	20	21	22
##	-9.188e-16	-4.652e-16	-7.122e-16	1.215e+01	-6.245e-15	2.708e-15	-2.068e-15
##	23	24	25	26	27	28	29
##	1.262e+00	4.208e-15	-2.559e+01	9.067e-15	2.433e+01	-1.381e-15	2.934e-15
##	30						
##	-2.944e-15						
##							
##	Coefficients: (1 not defined because of singularities)						
##		Estimate	Std. Error	t value	Pr(> t)		
##	(Intercept)	-102.808	184.182	-0.558	0.676		
##	x2	43.351	22.122	1.960	0.300		
##	x3	35.902	19.801	1.813	0.321		
##	x42.3	73.774	104.250	0.708	0.608		
##	x43	84.685	101.233	0.837	0.557		
##	x43.6	108.868	70.599	1.542	0.366		
##	x43.7	67.214	73.276	0.917	0.527		
##	x44.1	99.034	78.764	1.257	0.428		
##	x44.3	67.881	60.714	1.118	0.465		
##	x44.4	157.455	76.335	2.063	0.287		
##	x44.5	5.519	83.103	0.066	0.958		
##	x44.6	77.399	74.603	1.037	0.488		
##	x44.7	106.231	102.412	1.037	0.488		
##	x45.1	110.567	74.922	1.476	0.379		
##	x45.2	88.852	68.660	1.294	0.419		
##	x45.5	46.039	81.745	0.563	0.673		
##	x45.6	177.065	73.106	2.422	0.249		

## x45.9	148.094	84.770	1.747	0.331
## x46,2	65.725	78.118	0.841	0.555
## x46.1	41.271	95.130	0.434	0.739
## x46.2	140.653	79.258	1.775	0.327
## x46.7	22.644	116.450	0.194	0.878
## x46.8	160.618	109.261	1.470	0.380
## x47.4	72.708	97.642	0.745	0.593
## x47.6	218.845	93.809	2.333	0.258
## x5	20.655	20.613	1.002	0.499
## x8 0	-51.045	86.372	-0.591	0.660
## x81	33.319	52.651	0.633	0.641
## x8 1	NA	NA	NA	NA

##

Residual standard error: 45.53 on 1 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9928, Adjusted R-squared: 0.7996

F-statistic: 5.137 on 27 and 1 DF, p-value: 0.3374